Ship It: Fast, Small, Online

Module 10 · Zero to LLM

The kitchen

trick	kitchen	effect
KV cache	prep ahead	reuse past keys/values
quantisation	smaller pans	4–8× less memory
batching	batch orders	more users per GPU

Quantisation formats

int8 dynamic (CPU) · GPTQ / AWQ (4-bit GPU) · **GGUF** (llama.cpp, Ollama)

Always re-measure quality.

Where to deploy

option	cost model	control	ops
local / Ollama	hardware	full	you
cloud GPU container	per second	high	you
managed API (e.g. Grok)	per token	low	none
edge / browser	free	tiny models	low

Production checklist

rate limits · timeouts · responsible logging · filtering · cost alerts · **eval set before every change**

Next: M11 — what's next + capstone